

M002 · Pistons

Topic: Pistons and Mazes · **Course:** Maze Madness · **Level:** Advanced · **Time:** about 55 minutes

A **sticky piston** pulls a block back to open a path, but only while a lever keeps the signal ON. Flip the lever to power it, then cross the gap.

1 · Predict

Write what happens.

```
agent.move(FORWARD, 1)
agent.interact(...)
agent.move(FORWARD, 1)
```

The agent walks to a lever, flips it, and a sticky piston pulls a block back. Can the agent move forward now?


```
if not agent.detect(BLOCK, FORWARD):
    agent.interact(...)
else:
    agent.move(FORWARD, 1)
```

What does the agent do at a gap? What does it do on solid ground?

2 · Trace 🐾

```
while True:
    if not agent.detect(BLOCK, FORWARD):
        agent.interact(...)
        agent.move(FORWARD, 1)
    else:
        agent.move(FORWARD, 1)
```

Describe step by step what the agent does as it moves through the maze.

3 · Spot and Fix Bugs 🐛

Read the goal, write the fix, then explain it.

Bug A — Power the piston **first**, then cross the gap it opens.

```
agent.move(FORWARD, 1)
agent.move(FORWARD, 1)
agent.interact(...)
```

Fixed code:

Why was it wrong? Why does your fix work?

Bug B — `interact` should run only when there is a **gap** ahead. This flips a lever on every step.

```
while True:
    agent.interact(...)
    agent.move(FORWARD, 1)
```

Fixed code:

Why was it wrong? Why does your fix work?

Bug C — A sticky piston holds the bridge **only while the lever is ON**. The second `interact` turns it OFF and the bridge vanishes, so cross first.

```
agent.interact(...)  
agent.move(FORWARD, 1)  
agent.interact(...)  
agent.move(FORWARD, 1)
```

Fixed code:

Why was it wrong? Why does your fix work?

4 · Write Full Code 🖋️

Write code that walks the agent to the lever, flips it so the piston opens the path, crosses while the path is open, and reaches the goal. Use a chat trigger to start it.

```
def cross_gap(name, message):
    pass # your code here

player.on_chat("go", cross_gap)
```

5 · Show Your Work 📷 🎥

Open your homework world. Run your code so the agent crosses the piston gap and reaches the goal.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.